Fiscal Year:	FY 2021	**Task Last Updated:**	FY 05/28/2021
PI Name:	Ju, Yiguang Ph.D.		
Project Title:	Quantitative Studies of Cool Flame Transitions at Radiation/Stretch Extinction Using Counterflow Flames		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2015 Physical Sciences NNH15ZTT001N-15PSI-A: Use of the NASA Physical Sciences Informatics System – Appendix A
Start Date:	08/01/2016	**End Date:**	07/31/2021
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 7/31/2021 per NSSC information (Ed., 9/10/20) NOTE: End date changed to 7/31/2020 per NSSC information (Ed., 11/12/19) NOTE: Extended to 7/31/2019 per NSSC information (Ed., 5/7/19)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Reuter, Christopher M.S. (Princeton University)		
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Performance Goal No.:			
Performance Goal Text:			

Task Description:	Cool flames have significant implications for the control of engine knocking, ground and space fire safety, and combustion emissions. The recent International Space Station (ISS) combustion science experiments on flame extinguishment discovered the existence of cool flames for both large single droplets and droplet arrays after radiative extinction. Numerical simulations of droplet combustion also provide an explanation for the transition of hot flame to cool flame after radiative extinction. However, it remains unclear at what conditions the hot flames transfer to cool flames and how the cool flame flammable region is related to hot flame flammability limit and extinction limit. It is also not clear how oxygen concentration, dilution, pressure, diffusion, and convection transport processes affect the cool flame transition. More importantly, due to limits of space and payload, the ISS experiments did not provide detailed speciation and temperature distributions of cool flame structures for quantitative validation of kinetics and radiation model. By using the Flame Extinguishment (FLEX) experimental data from the Physical Sciences Informatics (PSI) system of combustion science, we propose to conduct new simulations and ground-based experiments using ozone-sensitized counterflow flames to * develop a new diagram of the flammability limits of diffusional cool flames affected by radiation, diffusion, and convection ; * bridge the knowledge gap between the flammability limits and extinction limits of hot flames and cool flames ; * predict and answer the question at what conditions the transition from hot flame extinction to cool flames is possible and verify the phenomena observed on the ISS flame extinguishment experiments ; * obtain new, quantitative data of extinction limit, flame temperature, and species distribution for the validation of kinetic and radiation models involving cool flames ; * extend the cool flame experiments from the original ISS droplet experiments to a more general ground-based combustion system involving radiation, diffusion, convection, and low-temperature chemistry while maintaining a one-dimensional geometry. The proposed new ground-based experiments and data analysis are complimentary to and not directly linked with the science objectives from the original ISS combustion science investigation, and will significantly advance the understanding of near-limit cool flame dynamics and contribute to the future control of engine knocking, ground and space fire safety, and combustion emissions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	An experimentally validated dimethyl ether (DME) and methane cool flame kinetic model was developed. The results showed that the existing kinetic model significantly over-predicts the cool flame extinction limits. However, by targeting the exothermic reactions of cool flames using the cool flame extinction limits, an updated chemical kinetic model for dimethyl ether/methane mixtures is developed and validated. This study shows the value of the ozone-assisted counterflow cool flame platform in examining the key reactions contributing to the heat release rate in cool flames. Diffusion warm flames of n-alkanes were successfully established. The warm flame of n-alkanes is dramatically influenced by the reactant concentrations and mixture temperatures. The primary cool flame stage is favored at lower temperatures, but the secondary warm flame stage is favored at elevated temperatures. Moreover, near extinction, the multistage warm flame collapses to a primary oxidation zone, governed by peroxy chemistry, whereas the secondary oxidation stage, governed by intermediate chemistry, is dominant near autoignition conditions. The results show that warm flames serve as a critical bridge between cool flames and hot flames. The counterflow cool flame extinction provides a good validation target and explanation for the NASA droplet cool flame extinction limit. In addition, the results benefit the development of efficient and low mission advanced engines for transportation.
Task Progress:	The extinction limits and cool flame radical index of non-premixed cool flames of large n-alkanes ranging from n-heptane to n-dodecane were measured in an atmospheric counterflow burner. The results demonstrate that an increase in fuel and oxygen concentration increases the cool flame extinction limit. In addition, with the increase in the carbon chain length of the n-alkane, the cool extinction limit also increases. The results also demonstrate that the dependence of cool flame extinction limit on ambient oxygen concentration for different n-alkanes is similar. The results are consistent with the measurements of extinction conditions in FLEX microgravity n-alkane-droplet experiments. The results also demonstrate that, due to the combined effects of multi-oxygen addition processes and the negative temperature coefficient effect, the cool flame low-temperature reactivity of n-alkanes is proportional to nth power of oxygen mole fraction, [O2]n. In this study, we measured n = 1.7, which is greater than n = 1.5 derived from the microgravity droplet experiment. The increase is due to the estimated NTC crossover temperature and radical dissociation being higher than the peak cool flame temperature predicted with detailed chemistry and transport effects. Furthermore, we also found that the transport-weighted enthalpy and radical index proposed for non-premixed hot flames are not directly applicable to the cool flames because of the existence of fuel and oxygen leakage in cool flames and the location of the flame with respect to the stagnation plane. By accounting for the cool flame thermal and mass transport effects with a mass-weighted enthalpy, a cool flame radical index was proposed to evaluate the cool flame reactivity and to correlate the extinction limits of large n-alkanes with different carbon lengths. The results showed that the cool flame extinction limits of n-alkanes collapse for the combination of the cool flame radical index and the mass-weighted enthalpy. In addition, the results showed that, with the increase of the chain length of the fuel molecule, the cool flame radical index increases. The derived scaling relation and radical index provide an alternative and convenient metric to evaluate low-temperature reactivity for new fuels and surrogate fuel mixtures. Further work to extend the current results to fuels with different moieties (and hence radical indices) is warranted and is of practical relevance.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
Articles in Peer-reviewed Journals	Zhou M, Yehia OR, Reuter CB, Burger CM, Murakami Y, Zhao H, Ju Y. "Kinetic effects of NO addition on n-dodecane cool and warm diffusion flames." Proceedings of the Combustion Institute. 2021;38(2):2351–60. https://doi.org/10.1016/j.proci.2020.06.002 , Jan-2021
Articles in Peer-reviewed Journals	Ju Y. "Understanding cool flames and warm flames." Proceedings of the Combustion Institute. 2021;38(1):83-119. https://doi.org/10.1016/j.proci.2020.09.019 , Jan-2021

Articles in Peer-reviewed Journals	Murakami Y, Reuter CB, Yehia OR, Ju Y. "Studies of autoignition-assisted nonpremixed cool flames." Proceedings of the Combustion Institute. 2021;38(2):2333–40. https://doi.org/10.1016/j.proci.2020.06.379 , Jan-2021
Articles in Peer-reviewed Journals	Zhou M, Yehia OR, Xu W, Reuter CB, Wang Z, Yan C, Jiang B, Ju Y. "The radical index and the effect of oxygen concentration on non-premixed cool flame extinction of large n-alkanes." Combustion and Flame. 2021 Sep;231:111471. Available online 5 May 2021. https://doi.org/10.1016/j.combustflame.2021.111471 , Sep-2021
Awards	Ju Y. (Yiguang Ju) "Plenary Lecturer, 38th International Symposium on Combustion, Adelaide, Australia, January 24-29, 2021. (Online)" Jan-2021
Awards	Ju Y. (Yiguang Ju) "2021 AIAA Propellants and Combustion Award, June 2021." Jun-2021